FEAT-ZD-48600 · FE-03 Identify the components/layers to be changed · Components, layers, sources or pipelines to change

Frontend agent · item task.components · source hobasa-frontend-agent/outputs/FEAT-ZD-48600/task.components.md · approved by automated checks and independent judge

task.components: FEAT-ZD-48600

Components and layers the Frontend lane will create or update for FEAT-ZD-48600 (ZD-48600, "Show which overdue invoices belong to vendors that usually pay late"). This is FE-03, attempt 1. The lane owns one task: TSK-07 / STORY-07, "Render the overdue risk table", in the ap_portal repository (feat.tasks, section 1; task.repo_access).

This step reads the existing code and lists the smallest set of changes that meets STORY-07. It changes no code; FE-04 writes the code. Statements marked **agent judgement** are this agent's proposals. No input states them as fact.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	FE-03 Identify the components/layers to be changed (attempt 1)	record:brief
Owner / gate	Developer / no gate: automated checks, the independent judge and sampled human evaluation	record:brief
Inputs	feat.tasks (b0fd4d2), feat.stories (0590ea4), feat.prd (908b58d), feat.trd (3637e51), feat.design (722a98f), task.repo_access (0097a2b)	record:brief
Task / story / HLD component	TSK-07 / STORY-07 / HC-8 (`renderOverdueRisk`)	feat.tasks (sections 1, 4); feat.design (A2)
Repository / branch	ap_portal, `feature/story` at `ff761ee`	task.repo_access
Code read	`src/ap_portal/index.js`, `tests/portal.test.js`, `package.json`, `ci.mjs`, `README.md`, `CLAUDE.md`	tool:repo; mcp:codegraph; file:ap_portal/ci.mjs; file:ap_portal/package.json; file:ap_portal/README.md; file:ap_portal/CLAUDE.md

1. What the existing code does

ap_portal has **one layer**: a stateless view-helper module that builds HTML strings. It has no routing, no state, no API client, no stylesheet and no build step. The checkout contains `CLAUDE.md`, `README.md`, `ci.mjs`, `package.json`, `src/ap_portal/index.js` and `tests/portal.test.js` (tool:repo; file:ap_portal/package.json).

Symbol in `src/ap_portal/index.js`	Line	What it does	Depended on by	Source		
Header comment	2	"AP portal view helpers: render the aging summary returned by the ap_aging service as HTML."	—	mcp:codegraph		

Symbol in src/ap_portal/index.js	Line	What it does	Depended on by	Source		
VERSION	3	'2.3.0' (also "version": "2.3.0" in package.json)	Exported	mcp:codegraph; file:ap_portal/package.json		
LABELS	4	Bucket key → display label	renderAgingSummary; exported	mcp:codegraph		
formatAmount(value)	6	`Number(value \	\	0).toFixed(2)`. This goes through a binary float, which conflicts with D-2026-044	renderAgingSummary; exported; existing test	mcp:codegraph; feat.trd (F-12); record:memory
renderAgingSummary(buckets)	10	<table class="aging-summary"> from template strings. It escapes nothing; its labels are constants	Exported; existing test	mcp:codegraph; feat.trd (F-13)		
module.exports	15	{ VERSION, LABELS, formatAmount, renderAgingSummary }	Tests via require('ap_portal')	mcp:codegraph; tool:repo		

Tests. tests/portal.test.js uses node:test and node:assert and loads the package with require('ap_portal'). It has two tests: formats amounts with two decimals and renders every bucket in order (tool:repo).

Note on codegraph. codegraph reports "no covering tests found" for formatAmount and LABELS (mcp:codegraph). Agent judgement: this is a false negative. Both are exercised by the existing tests, which load the module through require('ap_portal') with NODE_PATH: 'src'. The index most likely does not follow that resolution (file:ap_portal/ci.mjs; tool:repo).

CI binding. ci.mjs runs node --check on every .js file in src/ap_portal, then node --test with coverage of src/** and NODE_PATH: 'src'. It requires line coverage >= 90, at least one test run, no failures, no line matching console.log or eval(, and no line longer than 140 characters (file:ap_portal/ci.mjs).

2. Components to create or update (minimum change)

#	File / layer	Component	Kind	Change	Why it is needed	Source
C-FE-1	src/ap_portal/index.js (view helper)	RISK_CLASSES	New constant	['high', 'medium', 'low', 'unknown']	Allow-list for the row class (AC-07.3, AC-07.8)	feat.design (B4.2, B4.3); feat.trd (IF-7.4)
C-FE-2	same	OVERDUE_HEADINGS	New constant	['Invoice', 'Vendor', 'Days overdue', 'Amount']	Header row in order (AC-07.1, AC-07.2). Uses "Days overdue", never "Late"	feat.design (B4.2); feat.prd (NFR-10)
C-FE-3	same	escapeHtml(value)	New private helper	String(value).replace(/[&<>"']/g, …) with one lookup map	Escapes id and vendor (AC-07.7)	feat.design (B4.2, SE-2); feat.trd (IF-7.7)

#	File / layer	Component	Kind	Change	Why it is needed	Source
C-FE-4	same	`toTwoDecimalString(value)`	New private helper	String-only. Checks `typeof value === 'string'` and the pattern `^-?\d+(\.\d{1,2})?$`, pads the fraction with zeros to two digits, and throws `TypeError` otherwise	Shows the amount without `Number()`, `parseFloat` or `formatAmount` (AC-07.5, AC-07.6)	feat.design (B4.2, B4.4, DD-5, DD-8); feat.tasks (TSK-07 notes)
C-FE-5	same	`riskClass(risk)`	New private helper	`RISK_CLASSES.includes(risk) ? 'risk-' + risk : 'risk-unknown'`	Feed data can never inject a class name (AC-07.3, AC-07.8)	feat.design (B4.2); feat.trd (IF-7.4)
C-FE-6	same	`renderOverdueRisk(rows)`	New exported function	Returns `''` for `undefined`, `null` or `[]`. Throws `TypeError` for any other non-array value. Otherwise returns `<table class="overdue-risk">`, a plain header row, then one `<tr class="risk-…">` per row in the given order (id, vendor, days, `<td class="amount">` amount), then `</table>`. Throws `TypeError` when `days` fails `Number.isInteger`. No re-sorting, no Risk column or risk text, no row actions	HC-8; AC-07.1 to AC-07.4, AC-07.9	feat.design (B4.2 steps 1–4, B4.4); feat.tasks (TSK-07); feat.stories (AC-07.1 to AC-07.9)
C-FE-7	same	`module.exports` (line 15)	Updated	Add `renderOverdueRisk` and leave the existing four names unchanged. The helpers stay private	AC-07.10	feat.design (B4.2); feat.stories (AC-07.10); mcp:codegraph
C-FE-8	same	`VERSION` (line 3)	Updated	Next minor version. The lane confirms the number	DD-11; TK-07.5	feat.design (DD-11); feat.tasks (section 2)
C-FE-9	`package.json`	`"version"`	Updated	Same value as `VERSION`. Nothing else in the file changes	Keeps `VERSION` and `"version"` in step	feat.tasks (TSK-07 "Done when"); file:ap_portal/package.json

#	File / layer	Component	Kind	Change	Why it is needed	Source
C-FE-10	`tests/portal.test.js` (test layer)	New `node:test` cases	Added after the existing tests	The B4.5 tests plus the CL-FE6 coverage cases (see section 4)	TK-07.3; CI coverage of at least 90%	feat.design (B4.5); feat.tasks (TSK-07 "Tests to add")

How this keeps the change small (agent judgement)

1. **No new file.** All of the code goes into `src/ap_portal/index.js`. A new module would add a second `require` path, and `ci.mjs` would also have to compile it, so nothing is gained (feat.design DD-2; task.repo_access section 3; file:ap_portal/ci.mjs).
2. **Append, don't rewrite.** Put the two constants and the four functions between `renderAgingSummary` (ends at line 13) and `module.exports` (line 15). Lines 4–13 stay byte-for-byte the same (mcp:codegraph; feat.tasks TSK-07 "Must not change").
3. **Two existing lines change**: `VERSION` (line 3) and `module.exports` (line 15). The header comment on line 2 may be widened to mention the overdue risk table. That is optional; it is a comment only (mcp:codegraph).
4. **Build the header from** `OVERDUE_HEADINGS` with `map` and `join('')`. Follow the `renderAgingSummary` template-string style; no DOM or templating library (feat.design B4.2 step 4; DD-1).
5. **Keep the helpers small and private.** `riskClass` and `escapeHtml` are single expressions. The design names them as helpers, so they stay as functions rather than being inlined. This keeps the code traceable to B4.2 (feat.design B4.2).
6. **Don't reuse** `formatAmount`, even though it already gives two decimals. Reusing it would break D-2026-044 and fail T-14 (feat.trd F-12, R-2; record:memory).

Unchanged (must not change)

Item	Reason	Source
`LABELS`, `formatAmount`, `renderAgingSummary` and their two tests	Signature rule; AC-07.10; T-16	feat.tasks (TSK-07); feat.design (HC-9); file:ap_portal/CLAUDE.md
`ci.mjs`	The CI script is not in TSK-07's file budget	feat.tasks (TSK-07 "Files in budget")
`README.md`, `CLAUDE.md`	Not in the file budget	feat.tasks (TSK-07 "Files in budget")
Other repositories (ap_aging and the rest)	TSK-07 depends only on the CT-1 field list	feat.tasks (TSK-07 "Depends on"); task.repo_access (section 4)

3. Interface consumed: CT-1

`rows` is the result of `JSON.parse` on the ap_aging payload. It is an array of `{id: string, vendor: string, days: integer, amount: string, risk: string}`, already in the right order. `amount` is a two-decimal string with no exponent and no thousands separator. `risk` is one of `high`, `medium`, `low` or `unknown` (feat.design A3 CT-1, B4.3). This repository adds no transport (feat.design DD-3).

4. Branches the tests must cover (for the 90% line-coverage bar)

Branch in `renderOverdueRisk` / helpers	Test	Source
`undefined`, `null`, `[]` → `''`	`returns empty string for no rows`, plus `null`	feat.design (B4.5); feat.tasks (CL-FE6 list)
Non-array `rows` → `TypeError`	CL-FE6 case	feat.tasks (TSK-07)

Branch in `renderOverdueRisk` / helpers	Test	Source
Each of the four risk classes; an unexpected risk → `risk-unknown`	`maps every risk value to its class, unexpected risk renders risk-unknown`	feat.design (B4.5)
Amount unchanged; padded from `"980.5"`; non-string → `TypeError`; exponent, thousands separator or three decimals → `TypeError`	`shows amount strings unchanged, pads short amounts as strings, rejects non-string amounts`, CL-FE6 cases	feat.design (B4.5); feat.tasks (TSK-07)
Non-integer `days` → `TypeError`	CL-FE6 case	feat.tasks (TSK-07)
Escaping of <, &, " in `id` and `vendor`	`escapes id and vendor`	feat.design (B4.5)
Full table shape and row order	`renders overdue risk table`	feat.design (B4.5); feat.stories (AC-07.1)

5. Traceability

Component	Story AC	TRD tests
C-FE-1, C-FE-5	AC-07.3, AC-07.8	T-12, T-15
C-FE-2, C-FE-6	AC-07.1, AC-07.2, AC-07.4, AC-07.9	T-12, T-13
C-FE-3	AC-07.7	T-15
C-FE-4	AC-07.5, AC-07.6	T-14
C-FE-7, unchanged items	AC-07.10	T-16
C-FE-8, C-FE-9	TK-07.5	—

Sources: feat.stories (STORY-07); feat.trd (T-12 to T-16); feat.design (B4.5).

6. Assumptions (agent judgement)

ID	Assumption
CA-1	The helpers are not exported, as the `module.exports` list in feat.design B4.2 shows. Tests reach them through `renderOverdueRisk`
CA-2	The new code goes between `renderAgingSummary` and `module.exports`, so the diff touches no existing logic
CA-3	Styling for the `risk-*` and `overdue-risk` classes is outside this repository, which has no stylesheet. HTML class names are the only styling hook ap_portal provides

7. Open questions (all non-blocking)

ID	Question	Interim answer	Owner (proposed)
CMQ-1	Where are the `overdue-risk` and `risk-*` classes styled? ap_portal has no stylesheet, and the host page is not named (related to DQ-1)	Out of scope for TSK-07; the classes are emitted only	UX lead / Architect
CMQ-2	Which exact version number should the minor step use?	The lane confirms it at FE-04	Developer

Carried, unchanged: TQ-2 (`formatAmount` float), TQ-6, TQ-7, TQ-11 / OQ-B10 (padding), OQ-B8 (escaping), OQ-B9 and PQ-P7 (no Risk column or actions), DQ-3 (`TypeError` on contract violation), RAQ-1 (branch name `feature/story`). Each keeps the interim answer used above (feat.design D3; feat.stories; task.repo_access).

8. Validation check

Check	Result	Source
Existing code read and understood	Pass (section 1)	tool:repo; mcp:codegraph
Every component traces to the design and a story AC	Pass (sections 2, 5)	feat.design; feat.stories
Within the TSK-07 file budget (`index.js`, `package.json`, `tests/portal.test.js`)	Pass	feat.tasks (TSK-07)
Existing signatures kept	Pass	file:ap_portal/CLAUDE.md; feat.design (header "Architecture rule", A5 NF-4, D2)
D-2026-044 / L-011 (no float for money)	Consistent (C-FE-4; `formatAmount` not reused)	record:memory
Conflict with an approved output	None	—
Blocking questions	None	—